ASEAN
GEOAI FUSION 2026

TRAINING MODULE

Introduction to Machine Learning Concepts

Core ideas, model types, training, and deep learning foundations



What you'll be able to do

- **Describe** core concepts of machine learning and how models learn from data.
- **Identify** different types of machine learning and when to apply each.
- **Describe** considerations for training and evaluating machine learning models.
- **Describe** core concepts of deep learning and neural networks.

01	02	03	04
Core concepts	Model types	Training & evaluation	Deep learning

How the module is structured

- **1.0** Introduction — what machine learning is, where it comes from.
- **2.0** Machine learning models — features, labels, and the function y = f(x).
- **3.0** Types of ML — supervised (regression, classification) and unsupervised (clustering).
- **4.0** Deep learning — neural networks and layered functions.
- **5.0** Summary — key takeaways and what's next.

Module path

You are here → 1.0 Introduction

SECTION 1.0

Introduction

What machine learning is, where it came from, and why it matters.

What is machine learning?

ML sits at the intersection of two disciplines:

- **Data science** — discovering patterns and insights from data.
- **Software engineering** — building applications that solve problems.

The goal

Use data to create a predictive model that can be incorporated into a software application or service — a basis for most modern AI solutions.

ORIGIN

Statistics and mathematical modeling of data.

Use past observations to predict unknown outcomes or values.

WHY IT MATTERS

Foundational understanding for working with AI systems.

ML in everyday life

Three examples of ML turning data into predictions:

Retail

Ice Cream Sales

Combine historical sales with weather records to predict how many ice creams will sell on a given day, based on the forecast.

Healthcare

Diabetes Risk

Use clinical data from past patients to run automated tests predicting whether a new patient is at risk from diabetes — weight, blood glucose, and other metrics.

Research

Penguin Species

Automate identification of Adelie, Gentoo, or Chinstrap penguins from physical measurements like flipper length and bill width.

SECTION 2.0

Machine Learning Models

Models as software, training data, and the math of y = f(x).

Models are software that encode a function

Fundamentally, a machine learning model is:

A software application that encapsulates a function derived from data. The function takes new inputs and produces a prediction.

CORE NOTATION

$$y = f(x)$$

features → function → label

- **x** feature values — observed attributes of the thing being observed.
- **y** label — the value we want to predict.
- **f** function — calculation derived by the algorithm during training.

Training and inference — the two phases

TRAINING

- Use past observations as training data.
- Each observation has features (x) and a known label (y).
- An algorithm finds a relationship between x and y.
- The result is a model that encapsulates the function f.

INFERENCE

- Use the trained model on new observations.
- Provide new feature values (x) for unseen data.
- The model computes the prediction: $\hat{y} = f(x)$.
- Output is used by applications to make decisions.

In short: training learns the function; inference uses it.

SECTION 3.0

Types of Machine Learning Models

Supervised, unsupervised — and the problems each solves.

Supervised vs. unsupervised

SUPERVISED

Training data has features AND known labels.

- **Regression** — predict a numeric value.
- **Binary classification** — true / false, one of two classes.
- **Multiclass classification** — predict one of multiple classes.

Used to train models for known prediction targets.

UNSUPERVISED

Training data has only feature values — no labels.

- **Clustering** — group similar observations.
- **Discover structure** — classes, segments, anomalies.
- **Foundation** — can feed a later classification task.

Used when no ground-truth labels are available.

Predicting a numeric label

Regression is supervised ML where the label is a number.

Example use cases:

Ice cream sales	Temperature, rainfall, and wind-speed → number of units sold.
Property price	Size, bedrooms, and local socioeconomic metrics → selling price.
Fuel efficiency	Engine size, weight, width, height, length → miles per gallon.

Common thread: predict a continuous value from observed features.

Predicting a category, not a number

BINARY CLASSIFICATION

Predict one of two mutually exclusive classes.

- Diabetes risk from clinical metrics (weight, age, glucose).
- Loan default from income, credit history, age.
- Marketing response from demographics and past purchases.
- Output is a true / false or positive / negative decision.

MULTI-CLASS CLASSIFICATION

Predict one label from multiple possible classes.

- Penguin species: Adelie, Gentoo, or Chinstrap.
- Movie genre: comedy, horror, romance, adventure, sci-fi.
- Classes are mutually exclusive — a penguin is one species.
- Used when there is a known set of discrete categories.

Finding groups without labels

The most common form of unsupervised ML:

Identify similarities between observations based on their features and group them into discrete clusters.

EXAMPLES

- Group similar flowers by size, leaves, and petals.
- Identify customer segments from demographics and behavior.
- Often used to discover class structure before classification.

VS. CLASSIFICATION

Classification needs known classes in advance.

Clustering discovers the classes from the data itself.

In practice, clustering can define the classes used to train a later classification model.

Deep learning and key takeaways

DEEP LEARNING

- **Neural networks** — emulate biological neurons with math.
- **Layers** — multiple layers form a deeply nested function.
- **Deep Neural Networks (DNNs)** — the result of this layering.
- **Same idea:** y = f(x), but f is composed across many layers.

KEY TAKEAWAYS

- **ML** is the foundation of modern AI.
- **Models** are software that encode a function y = f(x).
- **Training** learns f; inference applies it to new data.
- **Supervised** uses labels; unsupervised discovers structure.
- **Deep learning** scales this idea with multi-layer networks.

End of module · Review the learning objectives before moving on.

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THANK YOU

Bootcamp and Hackathon for ASEAN Geospatial Innovation 2026